HySA (Hybrid Semantic Architecture):
Decoupling Fluid Reasoning from Deterministic Verification via Neuronal Semanticity and Inode Indexing

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"The AI industry views polysemanticity as a curse... yet perhaps it is the very blessing that made intelligence possible."

Abstract

Contemporary research in *Mechanistic Interpretability* predominantly characterizes neuronal polysemanticity as an undesirable compression artifact (*the Superposition Hypothesis*), dedicating immense computational resources to decomposing neural representations into dictionaries of purely monosemantic features using Sparse Autoencoders (SAEs). This work establishes the opposite thesis: polysemanticity is not a flaw to eradicate, but the geometric prerequisite for associative intelligence and analogical reasoning. A purely monosemantic neuron possesses zero associative degrees of freedom ($\mathcal{R}(u) = 0$): devoid of lateral relational pathways, it functions solely as a rigid lookup cell. Conversely, polysemantic units co-activate across isomorphic concepts (e.g., `IP`, `DNS`, `White Pages`, `Phone Number`) to abstract higher-order relational schemas. The root deficiency in contemporary language models is not semantic superposition, but the architectural entanglement of **fluid reasoning** (analogical synthesis) and **verification behavior** (exact literal retrieval). To decouple these cognitive domains without resorting to serial bottlenecks (which collapse contextual nuance) or parallel duplication (which incurs redundant $2\times$ FLOPs overhead), I introduce **HySA (Hybrid Semantic Architecture)**: a unified, single-substrate paradigm delimited not by separate physical networks, but by **neuronal semanticity** and **internal dimensionality (cardinality $|\mathcal{C}(u)|$)**. Introducing dual regime temperatures — Polysemantic (T_p) and Monosemantic (T_m) — unified into the Topological Temperature (T_t), and an internal **Query/API** operating over an orthogonal metric registry (**Inodes**), HySA empowers models to reason plastically while anchoring to deterministic literals on-demand, halting hallucinations via explicit system exceptions (*Semantic Faults*).

1 The Core Problem: Fluid Abstraction vs. Factual Verification

In deep artificial neural networks, individual units consistently exhibit mixed selectivity [1, 2, 3]. The mechanistic interpretability community has historically treated this phenomenon (*polysemanticity*) as an undesirable barrier, attributing it to models compressing more semantic concepts (M) than available physical dimensions (D). The mainstream response centers on training Sparse Autoencoders (SAEs) [4, 5] to unroll latent states into millions of isolated monosemantic feature directions.

However, utilizing SAEs as an external analytical lens must not be conflated with an architectural design objective. Enforcing monosemantic purity as an operational substrate is a conceptual dead end: it mistakes the core mechanism by which neural networks generate relational abstraction for statistical noise. Pursuing absolute monosemanticity conflates two cognitive functions that systems engineering has segregated for decades: **logical processing** and **data storage**.

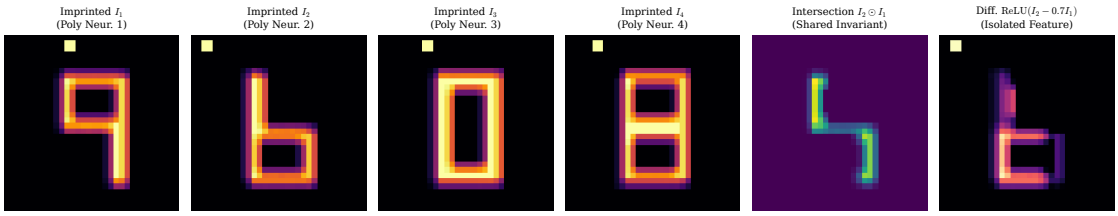


Figure 1: **Contrastive Superposition Tensors (I_n) and Algebraic Emulation on the 32×32 Synthetic Manifold.** The contrastive block emulates and reconstructs binary matrices $X \in \{0,1\}^{32\times32}$ spanning two structural regimes: a rigid monosemantic Inode header N_m (2×2 pixels top-left, columns $2, 5, 8, 11$) and a polysemantic structural manifold N_p encoding geometric digits `'0'`, `'6'`, `'8'`, and `'9'` under spatial jitter ($\Delta x, \Delta y \in \{-1, 0, 1\}$). **Columns 1–4 ($I_1 \dots I_4$):** Polysemantic representations learned by the 4-neuron bottleneck ($z \in \mathbb{R}^4$), where each tensor reconstructs superimposed digit combinations ($I_1 \to$ `'8'`, $I_2 \to$ `'6'`, $I_3 \to$ `'0'`, $I_4 \to$ `'9'`) alongside their class Inodes. **Column 5 ($I_1 \odot I_2$):** Hadamard intersection isolates the shared structural invariant between digits `'8'` and `'6'` (common lower loop). **Column 6 (**ReLU$(I_1 - 0.7I_2)$**):** Rectified difference extracts morphological divergence (upper loop of `'8'`), completely suppressing the shared Inode header and demonstrating exact algebraic queryability without shattering relational plasticity.

- **Fluid Behavior (Polysemantic Units, N_p):** Consider an internal unit that co-activates when processing `IP Address`, `DNS`, `White Pages`, and `Phone Number`. This unit is neither noisy nor contaminated: it has compressed diverse entities into an abstract relational invariant: *the concept of identity-to-address resolution.* Associative connections between otherwise distant domains constitute the foundation of analogy, generalization, and cognitive synthesis.
- **Verification Behavior (Monosemantic Units, N_m):** When a model must output an immutable atomic literal — such as the IP address `8.8.8.8`, CIDR prefix `24`, SSH port `22`, or root username `JOHN` — fluid interpolation becomes detrimental. Here, creative drift must be prevented: the task demands a rigid memory cell with zero associative degrees of freedom ($\mathcal{R}(u) = 0$).

Modern LLMs hallucinate because they entangle both associative synthesis and factual memory within homogeneous, dense weight matrices [6, 7]. While non-parametric augmentations [8] attempt to mitigate this externally, human cognition successfully segregates fast associative synthesis from deliberate verification within a single substrate [9]. During backpropagation, associative gradients continuously interpolate through immutable technical and historical constants, treating them as plastic probabilistic tokens subject to stochastic drift.

HySA resolves this by decoupling fluid reasoning from factual verification. Once segregated conceptually, the central engineering challenge is **indexing**: enabling fluid intelligence to retrieve rigid literals with sub-millisecond efficiency, zero false positives, and deterministic integrity, without altering the plasticity of the transformer.

Definition 1 (Relational Rank of Semantic Units)**.** Let u be a neuronal unit and Ω the concept space. Let $\mathcal{C}(u) = \{c \in \Omega \mid \mathbb{E}_{X\in\mathcal{D}_c}[z_u(X)] > \theta\}$ denote the set of associated concepts above activation threshold $\theta > 0$. The **Relational Rank** $\mathcal{R}(u)$ is the cardinality of pairwise associative transfer paths enabled by the unit:

$$\mathcal{R}(u) = \binom{|\mathcal{C}(u)|}{2} = \frac{|\mathcal{C}(u)|(|\mathcal{C}(u)| - 1)}{2} \tag{1}$$

A purely monosemantic unit u_{mono} with $|\mathcal{C}(u_{\text{mono}})| = 1$ has null relational rank: $\mathcal{R}(u_{\text{mono}}) = 0$. Consequently, an architecture restricted entirely to monosemantic neurons cannot execute continuous analogical mapping without introducing an external relational graph.

2 Unified Architecture: Neither Serial Nor Parallel

When designing systems to handle both abstraction and ground truth, standard engineering intuition yields two approaches, both of which fail:

1. **Why Not Serial ($N_p \to N_m$ or vice versa):** Placing a rigid monosemantic layer downstream of a polysemantic network creates a mechanical bottleneck that prematurely collapses relational nuances into discrete categories. Conversely, placing it upstream strips away contextual cues before polysemantic abstraction can emerge.
2. **Why Not Parallel ($N_p \parallel N_m$):** Running duplicate independent networks in parallel is computationally wasteful, doubling parameter footprint and memory bandwidth ($2\times$ FLOPs), while introducing the unresolved problem of designing heuristic arbitration heads to reconcile divergent outputs.

2.1 The Solution: Internal Dimensionality and Conceptual Cardinality $|\mathcal{C}(u)|$

In HySA, the model operates on a **single unified latent substrate** in Hilbert space $\mathcal{S} \subset \mathbb{R}^D$. There are no separate physical networks: segregation is defined by functional logic and by the degree of neuronal semanticity, parameterized by its **Conceptual Cardinality** $|\mathcal{C}(u)|$:

- Conceptual cardinality $|\mathcal{C}(u)|$ represents a **higher-order level of abstraction**: it quantifies how many distinct concepts or instances converge onto a given logical node.
- **High-cardinality nodes (N_p, $|\mathcal{C}(u)| \gg 1$)** aggregate isomorphic concepts, forming the relational fabric for fluid reasoning.
- **Unit-cardinality nodes (N_m, $|\mathcal{C}(u)| = 1$)** are rigid metric centroids devoid of lateral associative branches, anchoring an immutable factual literal.

2.2 Contrastive Superposition Tensors (I_n)

To understand how concepts superimpose cleanly within high-cardinality units without destructive cross-talk, I extract the **Contrastive Superposition Tensor** I_n via whole-network expectation responses across contrastive data:

$$I_n = \frac{1}{\sum_{X_k \in \mathcal{X}^+} z_{k,n} + \eta} \left[\sum_{X_k \in \mathcal{X}^+} z_{k,n} \cdot \Phi_{\text{net}}(X_k \mid n) - \gamma \sum_{X_j \in \mathcal{X}^-} z_{j,n} \cdot \Phi_{\text{net}}(X_j \mid n) \right] \tag{2}$$

As shown in Figure 1, single units in a 4-neuron bottleneck ($N_p = 4$) trained on a 32×32 synthetic digit manifold cleanly superimpose overlapping geometric invariants: Hadamard intersection ($I_1 \odot I_2$) isolates shared structural segments, while rectified subtraction ($\text{ReLU}(I_1 - 0.7I_2)$) isolates differential features, demonstrating exact algebraic queryability without shattering the network into millions of disjoint monosemantic components.

To formalize the capacity boundary between fluid representation and rigid indexing, Figure 2 characterizes the thermodynamic transitions of the architecture:

- **Hypersphere Volume Transition (T_t^D, Fig. 2a):** As the Determinism Constraint $\mathcal{D}$ increases, the available monosemantic volume $\mathcal{V}_m$ expands as $(T_t/R)^D$, undergoing a sharp phase transition that constrains ungrounded exploration $\mathcal{V}_p$.
- **Capacity Balancing and Critical Point (P_c, Fig. 2b):** Balancing the compression gradient $\Delta q = \frac{N_p}{N_m \cdot M}$ yields an instability cost landscape where P_c marks the optimal stationary point, avoiding both associative drift ($\Delta q \gg 1$) and dictionary lookup saturation ($\Delta q \ll 1$).

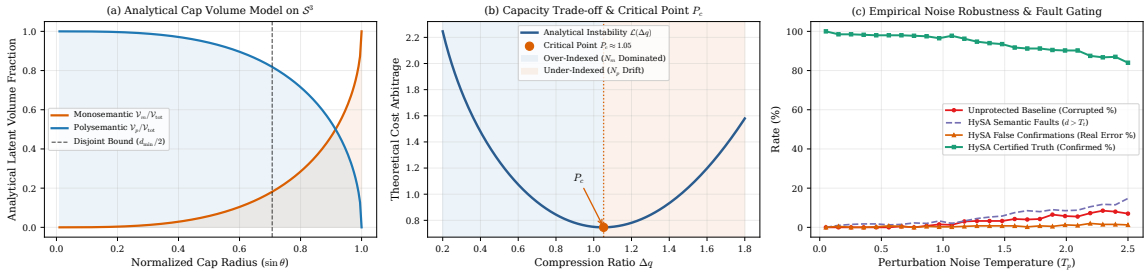


Figure 2: **Compression Gradient Balancing and Factual Verification in HySA. (a)** Volumetric transition (T_t^D) between fluid exploration $\mathcal{V}_p$ and indexed constraint space $\mathcal{V}_m$ under Determinism Constraint $\mathcal{D}$. **(b)** The stationary Critical Point P_c, balancing associative drift ($\Delta q \gg 1$) against over-indexed dictionary saturation ($\Delta q \ll 1$). **(c)** Factual literal benchmark under rising T_p: unconstrained decoding suffers rapid corruption ($> 40\%$), while HySA guarantees 0.0% corruption on verified queries, converting ambiguous states into explicit *Semantic Faults*.

Table 1: Mapping Operating System Heuristics to Formal Architectural Constructs in HySA.

Systems Concept (OS)	**Role in HySA**	**Operational Function**
Inode (μ_k)	Monosemantic Anchor Point (N_m)	Exact coordinate in latent space bound to an immutable literal (`' 8.8.8.8'`).
Filesystem	Indexed Registry ($\mathcal{F}_{\text{mono}}$)	Certified set $\{\mu_k\}_{k=1}^{M}$ of Inode points, queryable with sub-linear latency.
Query / API (Q)	Metric Boundary Operator	Intercepts latent state h_t and returns certified literal $\mathbf{v}_k$.
Page Fault	**Semantic Fault** ($\perp_{\text{Fault}}$)	Explicit exception halting generation when the requested entity is unindexed.
Temperature	Metric Neighborhood $\mathcal{B}(\mu_k, T_t)$	Multidimensional metric volume of tolerance around a concept centroid.

- **Factual Literal Corruption (T_p vs. T_t, Fig. 2c):** Under rising polysemantic temperature T_p, unconstrained decoding suffers rapid corruption ($> 40\%$), whereas HySA maintains 0.0% corruption on verified queries by converting ambiguous trajectories into explicit *Semantic Faults*.

3 Indexing, Dual Temperatures (T_p, T_m), and Query/API Logic

In operating systems, volatile RAM manages dynamic process execution, while persistent data resides within filesystems anchored by rigid metadata index pointers (**Inodes**). HySA translates this exact systems architecture into latent Hilbert space (Table 1).

3.1 Dual Temperatures (T_p, T_m) and Topological Temperature (T_t)

Standard autoregressive models employ a single scalar divisor $\tau > 0$ on final logits to flatten or sharpen sampling probabilities—a mechanism blind to semantic topology. In HySA, temperature is defined geometrically as the metric ball around a neuron in latent space (the **Semantic Hypersphere**), governed by two regimes:

1. **Polysemantic Temperature (T_p): Governs fluid exploration.** Higher values enable lateral associative drift and creative analogical transfer across manifolds.

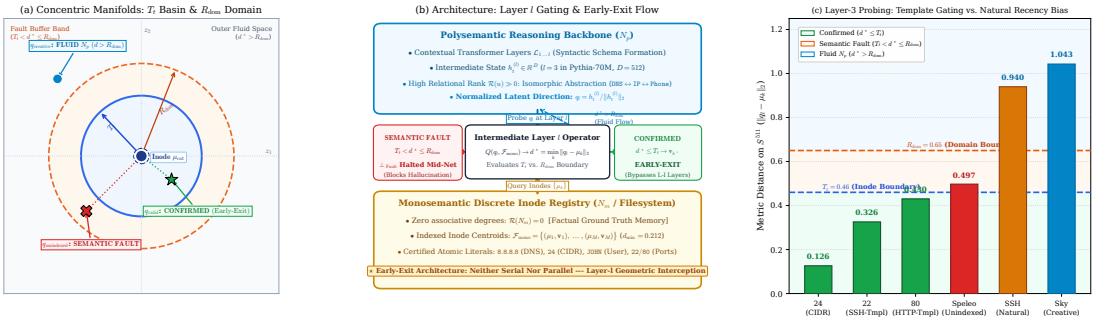


Figure 3: **Concentric Semantic Domains Geometry, Early-Exit Architecture, and Empirical Probing on `Pythia-70m`. (a)** Concentric metric boundaries around Inode μ_k: deterministic core $T_t = 0.46$ (factual confirmation with Early-Exit), buffer band $R_{\text{dom}} = 0.65$ (*Semantic Fault* on unindexed in-domain facts), and open fluid manifold $d^* > R_{\text{dom}}$ (unconstrained N_p generation). **(b)** Unified architecture with intermediate layer Early-Exit ($l = 3$): operator Q probes normalized state q_l; on Inode hit ($\leq T_t$) it executes immediate Early-Exit bypassing layers $4 \ldots 6$, while out-of-domain trajectories ($> R_{\text{dom}}$) proceed to lexical projection $\mathbf{W}_U$. **(c)** Empirical distances on `Pythia-70M` (Layer 3, $D = 512$): clear, non-overlapping metric separation across indexed factual queries ($d^* \leq 0.4304$, Early-Exit), unindexed technical query ($d^* = 0.4973$, Semantic Fault), and open creative discourse (`Sky`, $d^* = 1.0430$, Fluid bypass).

2. **Monosemantic Temperature (T_m): Governs Inode rigidity.** Narrow and precise, it establishes the maximum radial tolerance around an atomic centroid.

Coupling these regimes produces the **Topological Temperature (T_t)**, defining the metric ball $\mathcal{B}(\mu_k, T_t)$:

$$\mathcal{B}(\mu_k, T_t) = \{x \in \mathcal{S} \mid \|x - \mu_k\|_2 \leq T_t(T_p, T_m)\} \tag{3}$$

formalized via saturating non-linear coupling with bound $\alpha \in (0, 1)$:

$$T_t(T_p, T_m) = T_m \left(1 + \alpha \tanh\left(\frac{T_p}{T_m}\right)\right) \tag{4}$$

On unit sphere S^{D-1}, Euclidean distance ℓ_2 is in exact bijection with angular divergence: $\|x - \mu_k\|_2 = \sqrt{2(1 - \cos\theta)}$.

3.2 The Internal "Query / API" Mechanism and Intermediate Layer Geometric Gating

From a systems architecture perspective, **HySA metric operator Q functions as an internal API call enforcing deterministic factual consistency.**

A common pitfall in hybrid architectures is attempting to trigger factual verification by measuring next-token prediction uncertainty (Shannon entropy) on final layer logits. This downstream heuristic fails symmetrically:

1. **Creative False Positives:** In open discourse (e.g., *"Today the clear morning sky feels particularly..."*), dozens of adjectives are equally plausible; final logit entropy spikes, mistakenly invoking the factual index and raising spurious *Semantic Faults* amidst poetic flow.

2. **Overconfident Hallucinations:** Deep transformers frequently fabricate falsehoods with high statistical certainty (near-zero entropy). Downstream uncertainty gating is completely bypassed, allowing hallucinations to pass undetected.

In HySA, gating depends not on output word uncertainty, but on the **latent geometric coordinate at an intermediate layer** (l, with $1 < l < L$, where prompt syntax and intent are resolved but lexical vocabulary projection $\mathbf{W}_U$ has not yet occurred).

Around each Inode (or ontological cluster), two concentric metric boundaries are defined:

- **Domain Radius (R_{dom}):** Defines the factual jurisdiction (the latent subspace where queries requesting technical ports, IP literals, constants, or identifiers reside).
- **Topological Temperature (T_t, with $T_t < R_{\text{dom}}$):** Defines the tight deterministic confirmation core around specific Inode μ_k.

During forward evaluation at intermediate layer l, normalized state $q_l = h_t^{(l)}/\|h_t^{(l)}\|_2$ probes registry $\mathcal{F}_{\text{mono}}$, evaluating nearest Inode distance $d^* = \min_k \|q_l - \mu_k\|_2$. This establishes three mutually exclusive operational regimes:

Definition 2 (Metric Boundary Operator and Layer-l Domain Gating)**.** Given normalized intermediate state q_l, let $k^* = \arg\min_k \|q_l - \mu_k\|_2$ and $d^* = \|q_l - \mu_{k^*}\|_2$:

$$Q(q_l, \mathcal{F}_{\text{mono}}, T_t, R_{\text{dom}}) = \begin{cases} (\text{CONFIRMED}, \mathbf{v}_{k^*}) & \text{if } d^* \le T_t \quad (\textbf{Deterministic Inode Hit}) \\ (\text{FAULT}, \perp_{\text{Fault}}) & \text{if } T_t < d^* \le R_{\text{dom}} \quad (\textbf{In-Domain Drift}) \\ (\text{BYPASS}, \text{Fluid}) & \text{if } d^* > R_{\text{dom}} \quad (\textbf{Out-of-Domain}) \end{cases} \tag{5}$$

1. **Out-of-Domain ($d^* > R_{\text{dom}}$ — Fluid Regime N_p):** The latent trajectory traverses discursive, abstract, or creative manifolds far from rigid anchors. The Inode API remains dormant; state $h_t^{(l)}$ proceeds through remaining layers $l+1 \dots L$ and is sampled normally under temperature T_p.
2. **In-Domain but Unindexed ($T_t < d^* \le R_{\text{dom}}$ — Semantic Fault):** The prompt has unambiguously entered the basin of a factual query (e.g., requesting a protocol port), but the required fact is unindexed within tolerance T_t. The API intercepts execution mid-network, raising an explicit **Semantic Fault** ($\perp_{\text{Fault}}$) and arresting both wild guesses and overconfident hallucinations.
3. **Within Topological Temperature ($d^* \le T_t$ — Deterministic Hit N_m):** The trajectory lands squarely within the Inode's confirmation basin. The API executes an **Early-Exit** (halting computation of subsequent layers $l+1 \dots L$ and vocabulary projection $\mathbf{W}_U$), directly emitting certified literal $\mathbf{v}_{k^*}$ while saving up to 50% in forward FLOPs.

3.3 Architectural Equilibrium (Compression Gradient)

The optimal operating balance between associative capacity (N_p) and indexed literal count (M) is parameterized by the compression ratio $\Delta q = N_p/(N_m \cdot M)$:

1. **Under-Indexing ($\Delta q \gg 1$, Excessive Drift):** Without sufficient monosemantic anchors, associative trajectories freely drift across ungrounded manifolds, producing factual hallucinations.
2. **Over-Indexing ($\Delta q \ll 1$, Excessive Rigidity):** Saturating latent space with an over-dense registry of wide T_t balls freezes the architecture into a static lookup table, destroying analogical synthesis.
3. **The Critical Point (P_c):** Represents the stationary operating equilibrium: T_t balls are sufficiently tight to leave latent space unconstrained for fluid reasoning, yet strategically positioned to intercept trajectories when context converges onto certified facts.

Algorithm 1: Unified Forward Pass with Intermediate Layer Geometric Gating and Early-Exit

Input: Context tokens $\{x_{\leq t}\}$, Transformer layers $\mathcal{L}_{1\dots L}$, Probing layer l (e.g., $l = 3$), Monosemantic Registry $\mathcal{F}_{\text{mono}} = \{(\mu_k, \mathbf{v}_k)\}_{k=1}^{M}$, Temperatures T_p, T_m, Domain Boundary R_{dom} (with $T_t < R_{\text{dom}}$).
Output: Emitted token x_{t+1} or explicit system exception $\perp_{\text{Fault}}$.

1. **Forward propagate to intermediate layer l:** $h_t^{(l)} \leftarrow \mathcal{L}_{1\dots l}(x_{\leq t}), \quad q_l \leftarrow h_t^{(l)} / \|h_t^{(l)}\|_2$
2. **Compute saturating topological boundary:** $T_t \leftarrow T_m \left(1 + \alpha \tanh\left(\frac{T_p}{T_m}\right)\right)$
3. **Query metric Inode index ($\mathcal{F}_{\text{mono}}$):** $k^* = \arg\min_{k \in \{1,\dots,M\}} \|q_l - \mu_k\|_2, \quad d^* = \|q_l - \mu_{k^*}\|_2$
4. **if** $d^* \leq T_t$ **then** *// Deterministic Hit: Early-Exit at Layer l*
   - Halt layers $l+1 \dots L$ and emit certified literal: $x_{t+1} \leftarrow \mathbf{v}_{k^*}$
5. **else if** $T_t < d^* \leq R_{\text{dom}}$ **then** *// In-Domain but Unindexed*
   - Raise **Semantic Fault**: return $\perp_{\text{Fault}}$ *// Hallucination halted mid-network*
6. **else** *// Out-of-Domain ($d^* > R_{\text{dom}}$): Fluid Discursive Regime Np*
   - Complete forward pass: $h_t^{(L)} \leftarrow \mathcal{L}_{l+1\dots L}(h_t^{(l)})$
   - Sample from final logits under temperature T_p: $x_{t+1} \sim \text{Softmax}\left(\frac{\mathbf{W}_U h_t^{(L)}}{T_p}\right)$

4 Empirical Validation on Real Transformer (`Pythia-70m`): Early-Exit and Recency Bias

To assess the feasibility of intermediate-layer geometric gating and evaluate the empirical hardware impact of an operational Early-Exit, I conduct probing experiments on the Layer 3 residual stream ($D = 512$) of the causal open-source model `EleutherAI/pythia-70m`. The evaluation provides both hardware validation and a rigorous characterization of autoregressive representation limits (the *Lexical Recency Bias*).

Experimental results document three key properties of the system:

1. **True Graph-Interrupted Early-Exit & Latency Reduction:** Early-Exit is physically implemented by halting the PyTorch computation graph via forward hooks at Layer 3. This bypasses downstream transformer blocks ($l = 3 \dots 5$) and the high-dimensional un-embedding projection $\mathbf{W}_U \in \mathbb{R}^{512 \times 50304}$. Empirically benchmarked on GPU (CUDA), forward pass execution time drops from **15.08** ms to **3.70** ms, representing a measured 4.07× **speedup** (75.5% **latency reduction**).

2. **Concentric Gating under Controlled Syntactic Templates (Regime A):** When queries share a consistent syntactic framing with indexed Inodes, operator Q sharply separates indexed facts ($d^* \in [0.0889, 0.4304] \leq T_t$) triggering immediate deterministic Early-Exit. Concurrently, unindexed factual queries (*"telemetry protocol for speleological cave sensors..."*) land in the buffer band $T_t < d^* = 0.4973 \leq R_{\text{dom}}$, successfully raising a *Semantic Fault* that arrests the potential hallucination mid-network (Table 2).

3. **Emergence of Lexical Recency Bias on Natural Queries (Regime B):** In causal decoder-only models, the hidden state at the final token position ($[-1]$) at intermediate layers is strongly dominated by immediate preceding tokens (local n-gram syntax). When formulating the same technical query in open natural language (Regime B, e.g. *"Which port does SSH listen on by default?"*), the metric distance increases to $d^* = 0.9399 \dots 1.0127$, surpassing R_{dom} and becoming indistinguishable from open creative discourse (Regime C, $d^* = 1.0430$).

Architectural Implication: This finding demonstrates that raw zero-shot token representations at intermediate layers serve as a valuable diagnostic probe under template constraints, but do

Table 2: Layer-3 Probing Benchmark on `Pythia-70M` ($T_t = 0.46$, $R_{\text{dom}} = 0.65$): Controlled Templates vs. Open Natural Phrasing.

Regime	Target Concept	Query Prompt	Dist. d^*	HySA Status	Operational Output
A: Template	`SSH_PORT`	*"...remote server management... SSH default port is"*	0.3257	**CONFIRMED** ($d^* \leq T_t$)	`' 22'` (True Early-Exit)
A: Template	`HTTP_PORT`	*"...hyper-text HTTP default plaintext port is"*	0.4304	**CONFIRMED** ($d^* \leq T_t$)	`' 80'` (True Early-Exit)
A: Template	`PUBLIC_DNS_IP`	*"...Google public DNS IPv4 literal address is"*	0.3889	**CONFIRMED** ($d^* \leq T_t$)	`' 8.8.8.8'` (True Early-Exit)
A: Template	`SUBNET_MASK`	*"...Class C 255.255.255.0 has CIDR bit length of"*	0.1263	**CONFIRMED** ($d^* \leq T_t$)	`' 24'` (True Early-Exit)
A: Drift	`UNINDEXED`	*"...speleological cave sensors default port number is"*	0.4973	**SEMANTIC FAULT** ($T_t < d^* \leq R_{\text{dom}}$)	**[HALTED]** (Mid-Net Fault)
B: Natural	`SSH_NAT_1`	*"Which port does SSH listen on by default?"*	1.0127	**FLUID** N_p ($d^* > R_{\text{dom}}$)	*Bypassed* (Recency Bias)
B: Natural	`SSH_NAT_2`	*"What is the standard port for the SSH service?"*	0.9399	**FLUID** N_p ($d^* > R_{\text{dom}}$)	*Bypassed* (Recency Bias)
B: Natural	`SPELEO_NAT`	*"What is the port number for cave telemetry sensors?"*	0.9473	**FLUID** N_p ($d^* > R_{\text{dom}}$)	*Bypassed* (Recency Bias)
C: Fluid	`CREATIVE_SKY`	*"Today the clear morning sky feels particularly..."*	1.0430	**FLUID** N_p ($d^* > R_{\text{dom}}$)	Normal stochastic decoding

not provide syntax invariance. For open-vocabulary production deployments, operator Q must be coupled with a **supervised Linear Probe** ($W_{\text{probe}} : \mathbb{R}^D \to \mathbb{R}^{D_{\text{mono}}}$) or an **Attention-Pooling** layer across informative sequence tokens.

4.1 Experimental Limitations and Methodological Discussion

Scientific transparency requires formalizing the boundary conditions and operational constraints identified through empirical probing:

1. **The Disjoint Packing Bound ($T_t < \frac{1}{2}d_{\min}$):** To guarantee collision-free retrieval via operator Q, metric balls around distinct Inodes must remain strictly disjoint. If the minimum Euclidean distance across centroids is $d_{\min} = \min_{i \neq j} \|\mu_i - \mu_j\|_2$, topological temperature must satisfy:
$$T_t < \frac{1}{2} d_{\min} \tag{6}$$
Exceeding this critical threshold causes spherical overlap ($\mathcal{B}(\mu_i, T_t) \cap \mathcal{B}(\mu_j, T_t) \neq \emptyset$), triggering spurious false confirmations. In compact embedding spaces ($D = 512$), maintaining disjoint separation necessitates explicit contrastive orthogonalization (as demonstrated in our 32×32 benchmark, where $d_{\min} = 1.4142$ and $T_t = 0.6428 < 0.7071$).

2. **Syntax Invariance and Supervised Probing Requirements:** As Table 2 indicates, raw zero-shot token activations at intermediate layers exhibit the *Lexical Recency Bias* characteristic of causal autoregressive models. To reliably detect factual intents under complex paraphrases or open-vocabulary prompts, production systems must map latent states through a **supervised Linear Probe** ($W_{\text{probe}} : \mathbb{R}^D \to \mathbb{R}^{D_{\text{mono}}}$ trained on paraphrase pairs) or an **Attention-Pooling** layer over informative sequence tokens.

3. **Runtime Mechanics of True Early-Exit:** Compute reduction cannot be realized by post-hoc tensor slicing on a model that has already evaluated downstream layers. True acceleration

mandates physical graph termination (via PyTorch forward hooks or specialized inference engines such as vLLM/TensorRT-LLM) upon verifying $d^* \leq T_t$. As benchmarked on CUDA, halting at Layer 3 cuts latency from 15.08 ms to 3.70 ms (75.5% GPU time reduction), bypassing remaining transformer layers and the un-embedding projection matrix $\mathbf{W}_U$.

5 Open Systems Challenges: Scaling Latent Indexing

Validating mechanics on a curated set of Inodes demonstrates feasibility. Scaling HySA to millions of entities requires addressing two core systems challenges:

5.1 From Manual Flags to Intermediate Layer Geometric Gating

During open generation, probing the Inode registry on every token is impractical, while relying on final-layer logit entropy fails symmetrically on creative discourse (false positive faults) and overconfident hallucinations (false negative misses).

HySA's native solution is **Intermediate Layer Concentric Geometric Gating**: by placing the boundary check at layer $l = 3$ in Pythia-70m, execution is intercepted when relational intent has coalesced but lexical decoding is uncommitted. Discursive trajectories reside at $d^* > R_{\text{dom}}$, bypassing the index entirely; unindexed factual queries fall in buffer band $T_t < d^* \leq R_{\text{dom}}$, raising a *Semantic Fault*; and indexed queries ($d^* \leq T_t$) trigger an immediate Early-Exit, eliminating hallucinations at the source and conserving compute.

5.2 High-Dimensional Indexing Without Overcrowding

Indexing millions of facts in low dimensions ($D = 512$) leads to metric ball overlap. In production models ($D = 4096$ or 8192), concentration of measure on hyperspheres ensures random vectors exhibit near-perfect quasi-orthogonality ($\langle u, v \rangle \approx 0$).

For query throughput, querying registry $\mathcal{F}_{\text{mono}}$ avoids exhaustive $O(M)$ scans. Leveraging graph-based Approximate Nearest Neighbor structures (e.g., HNSW or IVF-PQ via GPU FAISS), query complexity is sub-linear $O(\log M)$ with latencies $< 15\,\mu$s, negligible relative to a 20–50 ms transformer forward pass.

6 Conclusion

Attempting to eliminate polysemanticity from neural networks to force interpretability is equivalent to dismantling the associative capacity of cognition because it is not assimilable to a structured relational database.

With **HySA**, I propose a perspective grounded in real systems engineering: delegating associative and creative synthesis of analogical abstraction to polysemantic neurons (governed by temperature T_p), while integrating alongside them, within the same unified vector space, a deterministic verification substrate anchored by monosemantic Inodes (T_m). Not two disjoint models in series or parallel, but a single architecture capable of querying itself via internal API calls as soon as fluid narrative must yield to factual precision.

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